

Total Body Weight is Not a Significant Predictor of Theophylline Elimination Half-Life: A Retrospective Analysis

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Abstract

Theophylline is a bronchodilator with a narrow therapeutic index, and its clearance is highly variable between individuals. Consequently, ensuring proper loading and maintenance doses is essential to reduce common and severe side effects. Half-life measurements are a crucial pharmacological metric that aids physicians in correctly administering pharmaceuticals with such narrow indexes like Theophylline. While the metabolic factors affecting theophylline clearance from the body are known, little work has been done to determine the effect patient total body weight has on the elimination half-life. Therefore, the goal of this retrospective analysis is to determine if total body weight is a significant predictor of the elimination half-life of theophylline. Using the publicly available Theoph dataset ($N = 12$), elimination half-lives were derived from log-transformed concentration-time curves. A simple linear regression revealed that total body weight did not significantly predict half life ($p = 0.404$, $R^2 = 0.07$). As a result, total body weight is an unreliable metric for predicting theophylline clearance or guiding maintenance dosing. The lack of correlation is likely due to the drug's distribution primarily into lean mass and its highly variable hepatic metabolism via cytochrome P450 enzymes.

Introduction

Although theophylline has been used as a bronchodilator to treat airway diseases for decades, its clinical utility is limited by a narrow therapeutic index where the required dosages frequently cause side effects ranging from nausea, vomiting, and headaches to more serious complications like cardiac arrhythmias and seizures (Barnes, 2010; Hendeles & Weinberger, 1980; Journey & Bentley, 2026). Understanding how theophylline is eliminated from the body is important for balancing these side effects with therapeutic benefits, as side effects are directly related to theophylline plasma concentration (Barnes, 2010; Stavric, 1988). Elimination half-life is a crucial pharmacokinetic measure that defines how long it takes for a drug's plasma concentration to decrease by half, a rate that can vary dramatically among individual patients (Cheymol, 2000; Hendeles & Weinberger, 1980).

Individual differences in the clearance of theophylline can be attributed to factors that influence hepatic metabolism such as smoking, liver disease, old age, and certain dietary choices (Barnes, 2010). However, it is unclear whether patient specific metrics like total body weight influence elimination half-life of theophylline. It is logically assumed that an increase in body mass may alter theophylline's volume of distribution or its clearance rate, thereby impacting its half-life.

Therefore, the purpose of this retrospective analysis is to evaluate the relationship between patient body weight and theophylline elimination half-life using the publicly available Theoph dataset (Boeckmann et al., 1994). I tested the null hypothesis that there is no linear relationship between a patient's total body weight and their theophylline elimination half-life, against the alternative hypothesis that a significant linear relationship exists.

Methods

A retrospective, secondary analysis was conducted using the publicly available Theoph dataset in R Studio (Boeckmann et al., 1994). The original data was collected by Dr. Robert Upton around 1980 and contains pharmacokinetic data from $N = 12$ subjects who received a single oral dose of theophylline. Unfortunately, it is unclear if Dr. Upton published this data as the original manuscript could not be located.

The predictor variable was patient body weight (kg), and the primary outcome measure was the theophylline elimination half-life (hours). To derive the half-life for each subject, concentration data from the initial absorption phase were excluded to isolate the elimination phase. The remaining theophylline concentrations were log-transformed ($\ln(\text{conc})$) to linearize the elimination phase. The elimination rate constant (K_{el}) was derived by calculating the absolute value of the slope from a simple linear regression of log-concentration plotted against time for each individual subject (Figure S1). The elimination half-life ($t_{1/2}$) was calculated using the standard pharmacokinetic formula:

$$t_{1/2} = \frac{\ln(2)}{K_{el}}$$

To test the hypothesis that patient body weight correlates with elimination half-life of theophylline, a simple linear regression (Ordinary Least Squares) model was used. All data processing, subsetting, and statistical analyses were conducted in RStudio. The threshold for statistical significance was set at $\alpha = 0.05$.

Results

The calculated theophylline elimination half-lives for the 12 subjects ranged from 6.16 to 14.3 hours, with baseline body weights ranging from 54.6 to 86.4 kg. A simple linear regression was conducted to determine if total body weight is a significant predictor of theophylline elimination half-life (Figure 1). The model indicated that body weight did not significantly predict half-life ($p = 0.404$), explaining only a negligible proportion of the variance in clearance rates ($R^2 = 0.07$) (Table 1). The absence of a significant linear relationship between the two variables is illustrated in Figure 1.

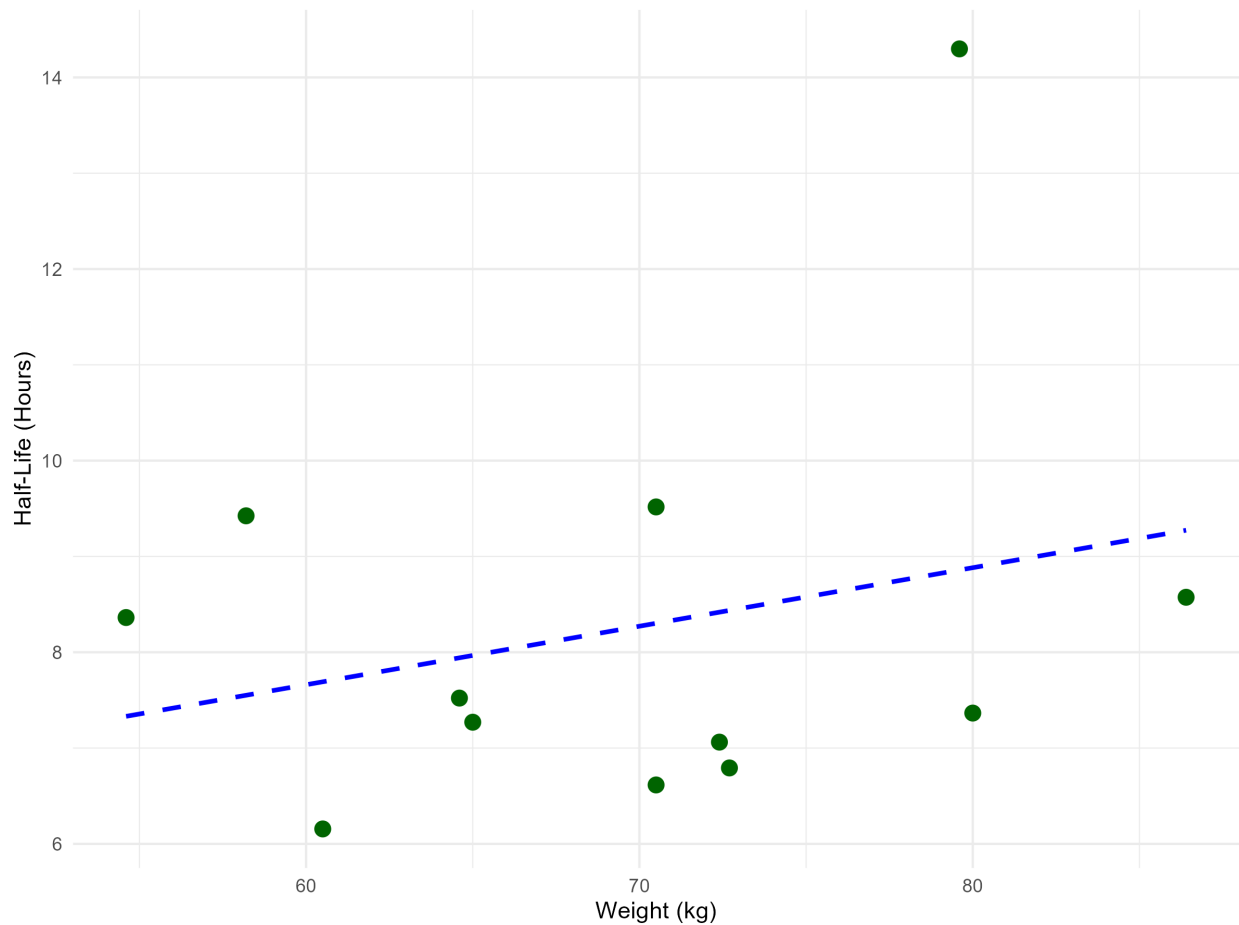


Figure 1: Relationship between total body weight (kg) and calculated theophylline elimination half-life (hours) in $N = 12$ subjects following a single oral dose. The blue dashed line represents the line of best fit ($y = 0.061x + 3.996$; $R^2 = 0.07$), highlighting the lack of a significant linear relationship ($p = 0.404$).

Table 1: Linear regression predicting theophylline elimination half-life from total body weight.

Predictor	Estimate	Std. Error	t-value	p-value
Intercept	3.996	4.921	0.812	0.436
Total Body Weight (kg)	0.061	0.070	0.871	0.404

Discussion

Based on the analysis, I fail to reject the null hypothesis that there is no linear relationship between a patient’s total body mass (kg) and their theophylline elimination half-life ($p = 0.404$, $R^2 = 0.07$). The average elimination half-life of theophylline in healthy, non-smoking adults is around 8 hours (Journey & Bentley, 2026). However, individual variability in theophylline half-life ranges from 3-4 hours to more than 24 hours in patients with liver disease, heart failure, or severe viral illness (Hendeles & Weinberger, 1980). Despite great individual variability in the half-life of theophylline, I can conclude that patient total body mass (kg) is a poor and unreliable predictor of how fast theophylline is eliminated from a patient’s system and should not be used as a proxy to calculate maintenance dosages (Figure 1).

This finding can be explained by evidence that theophylline is a hydrophilic/water-soluble drug, which means theophylline will distribute into lean body mass and extracellular fluid rather than fatty tissues (Cheymol, 2000). Therefore, using total body mass to predict half-life is not recommended, as it will introduce noise by including the mass of fatty tissue. Therefore, a high total body mass due to higher fatty tissue content does not necessarily display differences in half-life for that reason alone.

Additional metabolic evidence can explain why total body mass is a poor predictor of theophylline half-life. Theophylline is heavily metabolized by liver enzymes, namely cytochrome P450 (Barnes, 2010; Hendeles & Weinberger, 1980; Journey & Bentley, 2026). Factors that interact with this enzyme, like smoking, age, heart failure, and liver dysfunction, are known to greatly affect the half-life of theophylline (Hendeles & Weinberger, 1980).

Since theophylline is not readily distributed into fatty tissue and fatty tissue percent does not directly affect cytochrome P450 or liver metabolism, it is a poor predictor of elimination half-life. The Theoph dataset only provides variables relating to total body weight, dose administered, and serum concentrations at different time points after administration; therefore, the dataset lacked variables for lean body mass, smoking status, and heart or liver conditions that may have contributed to the individual variability in half-life (Figure 1).

Relying on total body mass for theophylline dosing is dangerous due to its narrow therapeutic index (Hendeles & Weinberger, 1980; Journey & Bentley, 2026). Future models and datasets reporting pharmacokinetic measures are recommended to include as much information as possible to accurately predict elimination half-life and prevent toxicity.

Data Availability and Reproducibility

Data and code are publicly available in the GitHub repository for this project (https://github.com/sammesenekal/senekal_mini-project).

The pre-registration and data management plan are available on the OSF for this project (https://osf.io/as843/overview?view_only=0386ce0964aa458194d05509746b3df0)

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Supplementary Materials

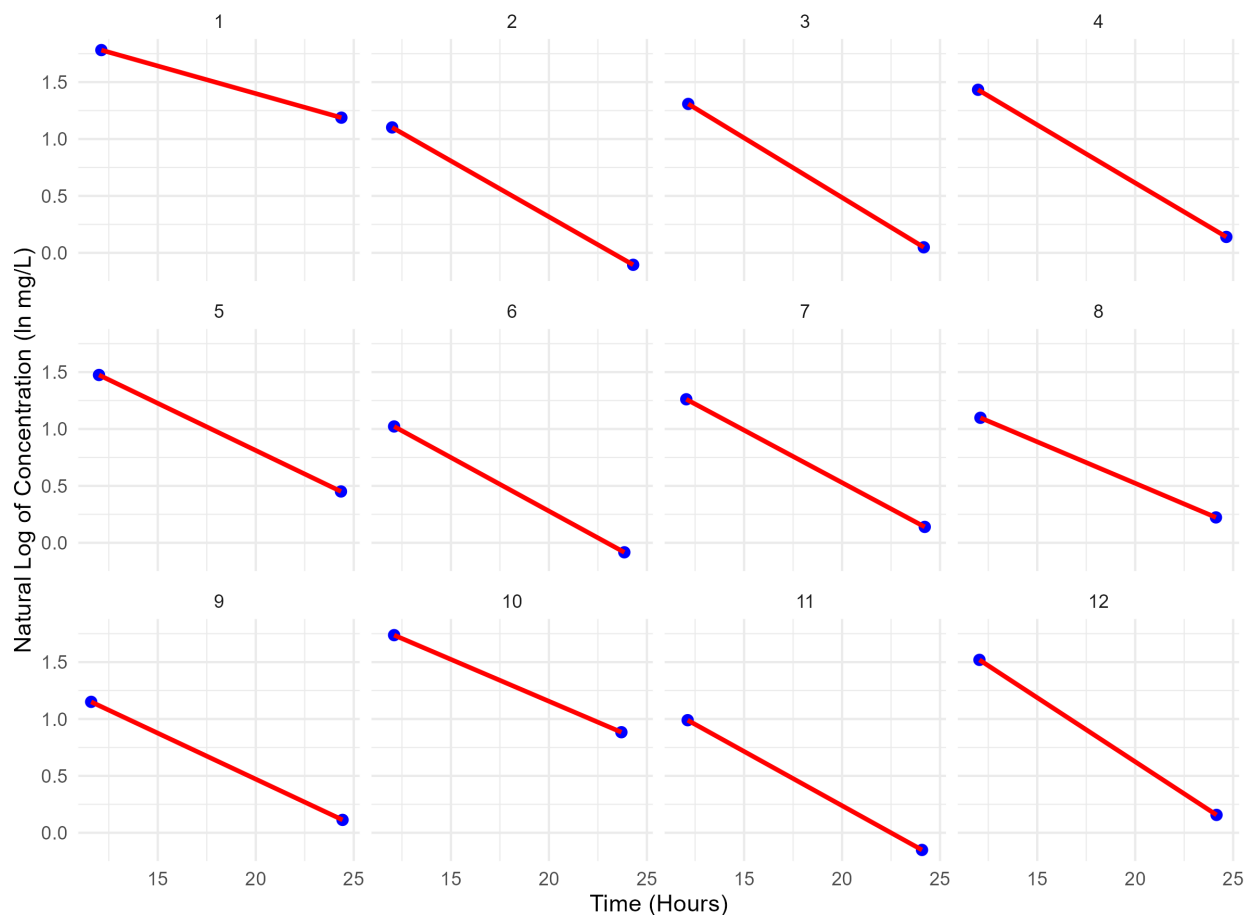


Figure S1: Individual theophylline log-concentration time curves for subjects 1-12. The blue dots represent the individual blood concentration measurements taken for each person during the elimination phase. The red solid lines represent the terminal elimination phase used for the calculation of the elimination rate constant (K_{el}).